Beyond Binary: Unified Detection and Attribution of AI-Generated Code via Hierarchical Family-Aware Learning

Research Proposal — NeurIPS 2026

Student Name

University Name

April 1, 2026

Outline

1. Problem: Why Binary Detection is Not Enough
2. Benchmarks: Three Complementary Challenges
3. Method 1: SpectralCode (Completed)
4. Results: SpectralCode (Completed)
5. Method 2: HierTreeCode (In Progress)
6. Evaluation Plan & Timeline
7. Experiment Insights

Binary Detection is Solved — The Real Challenges Remain

Binary: Human vs Machine (CoDET-M4)

Method	F1 (%)
SVM	72.19
CatBoost	88.78
CodeBERT	95.70
CodeT5	98.35
UniXcoder	98.65
SpectralCode (ours)	**99.06**

Three Unsolved Challenges

Authorship: *Which* LLM generated it?
(6–12 classes, best SOTA only 66% F1)

OOD: Train Python → test Go?
(99% → 30% F1 collapse on AICD)

Adversarial: DPO-tuned mimicry?
(Models fine-tuned to write like humans)

⚠ >99% F1 ⇒ task is saturated

Our vision: A unified method evaluated across **3 benchmarks**, **3.6M samples**, covering binary + attribution + OOD + adversarial

The Bigger Picture: From Detection to Understanding

The Evolution of the Threat

2023 "Is this code written by AI?" → Binary detection (**solved**: >99% F1)

2024 "Which model family generated it?" → Attribution (**hard**: best 66% F1, 6-class)

2025 "Can the model fool the detector?" → DPO-tuned adversarial code (DROID, EMNLP'25)

2026 "Does it generalize to unseen languages?" → OOD collapse: 99% → 30% (AICD, EACL'26)

Why Current Methods Fail at Attribution

- LLMs form **family trees**: Nxcode = fine-tuned Qwen1.5, CodeLlama = fine-tuned Llama
- Fine-tuned models inherit parent's "stylometric DNA" — nearly identical coding patterns
- Standard classifiers treat all classes as **equidistant** → 40%+ confusion between parent-child pairs
- Qwen1.5 F1 = 0.41 (near random for 6-class)

Our Key Insight

The family tree of LLMs is not a nuisance — it is **structured prior knowledge**.

Models within the same family should be *closer but separable* in embedding space.

⇒ **Encode genealogy as distance constraints**

Three Benchmarks — Why These Three?

	CoDET-M4 Orel et al., ACL Findings 2025	**DroidCollection** Orel et al., EMNLP 2025	**AICDBench** Orel et al., EACL 2026
Scale	500K samples	1.06M samples	2.05M samples
Languages	3 (Py, Java, C++)	7 languages	9 languages
Models	5 generators	43 models, 11 families	77 models, 11 families
Sources	CF, GitHub, LeetCode	General, Algo, Research	Algo, Research, General
Binary	✓ (saturated 99%)	✓ (saturated 97%)	✓ + **4-split OOD**
Attribution	✓ 6-class	—	✓ 12-class family
Refined	✗	✓ **3 scenarios**	✓ (hybrid class)
Adversarial	✗	✓ **DPO 157K pairs**	✓ (in Task 3)

CoDET-M4: Attribution

Best for fine-grained authorship (6-class). Known Nxcode↔Qwen confusion. Per-lang/source breakdowns.

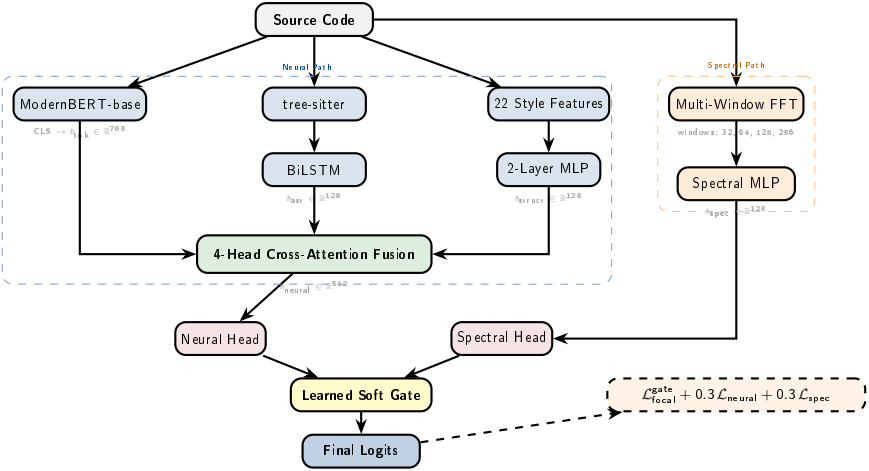
DROID: Adversarial

Machine-refined (continue, gap-fill, rewrite). DPO evasion attacks. Varied decoding strategies (temp, top-k, beam).

AICD: OOD Stress-Test

77 models incl. reasoning LLMs. Train 3 lang → test 9 lang. Progressive 4-split OOD evaluation.

SpectralCode — Architecture Pipeline



SpectralCode — What Makes Each Stream Novel?

Neural Path

Token: ModernBERT-base (SDPA)
→ CLS token, semantic+lexical

AST: tree-sitter → 67 node types → BiLSTM
→ syntactic structure

Structural: 22 features — indent patterns, naming style (snake_case/camelCase ratio), comment density, complexity counts

Fusion: 4-head cross-attention + gated projection → $h_{\text{neural}} \in \mathbb{R}^{512}$

Spectral Path (Novel)

Key insight: LLM decoding creates *frequency artifacts* — periodic patterns from autoregressive sampling (top-k, top-p, temperature).

Process:
Token IDs → 1D signal → FFT at 4 windows (32, 64, 128, 256)

Per window (16 features):

- 8 band energies
- Spectral centroid
- Rolloff, flatness, peak freq

→ 64-dim → MLP → $h_{\text{spec}} \in \mathbb{R}^{128}$

Inspired by deepfake spectral forensics (Frank et al., ICML 2020)

Learned Gate

Why? Not all code benefits equally from spectral vs neural analysis.

Gate: MLP([neural; spec]) → softmax weights (w_n, w_s)

$$\text{logits} = w_n \cdot f_n(h_{\text{neural}}) + w_s \cdot f_s(h_{\text{spec}})$$

Auxiliary losses on each head individually ensure both paths learn useful representations.

Total:
$\mathcal{L}_{\text{focal}} + 0.3\,\mathcal{L}_{\text{neural}} + 0.3\,\mathcal{L}_{\text{spec}}$

CoDET-M4: SpectralCode vs Paper Baselines (ACL Findings 2025)

Binary Detection (Macro-F1 %)

	P	R	F1	Acc
CatBoost	88.7	88.8	88.8	88.8
CodeBERT	95.7	95.7	95.7	95.7
CodeT5	98.4	98.4	98.4	98.4
UniXcoder	98.7	98.7	98.7	98.7
Ours	**99.1**	**99.1**	**99.1**	**99.1**

Author Attribution (6-class F1)

	F1	Δ
SVM	27.63	—
CatBoost	45.42	—
CodeT5	62.45	—
CodeBERT	64.80	—
UniXcoder	66.33	—
SpectralCode	**69.82**	+3.49

Per-Language Author F1 (%)

	C++	Java	Python
SpectralCode	72.31	70.12	69.22

Per-Source Author F1 (%)

	CF	GitHub	LC
SpectralCode	77.17	56.18	60.35

GitHub is the main bottleneck (↓21% vs CF)

DroidCollection: SpectralCode **Beats** DroidDetect-Base (EMNLP 2025)

Comparison with DROID Paper (Weighted-F1 %)

Type	Method	3-cls	4-cls
Zero-shot	Fast-DetectGPT	64.54	—
	GPTZero	49.10	—
Fine-tuned	CatBoost	72.31	—
	GCN	51.63	—
Full train	GPT-SnifferFT	82.75	—
	CoDet-M4FT	83.25	—
	DroidDetect-B	86.76	92.95
	DroidDetect-L	**88.78**	**94.30**
Ours	**SpectralCode**	**88.77**	**88.02**

All numbers: weighted-F1. DROID paper: domain-avg. Ours: overall.

Key Findings

- ✓ **Beats DroidDetect-Base** (88.77 > 86.76)
- ✓ **Ties DroidDetect-Large** (88.77 ≈ 88.78)
- ✓ Beats **all** zero-shot & fine-tuned
- → Using **base** model (149M) vs paper **large** (395M)!

Scaling Potential

Currently trained on only **100K / 1.06M** (∼1/10) of DROID data.

Full data training expected to improve further.

AICDBench: Ongoing Exploration (EACL 2026)

AICD Paper Baselines vs SpectralCode (Macro-F1 %)

Method	T1 (2-cls)	T2 (12-cls)	T3 (4-cls)
CodeBERT	28.64	23.71	55.60
UniXcoder	25.51	26.64	54.21
DeBERTa	34.13	12.08	54.65
ModernBERT	**30.61**	**32.84**	**61.65**
SpectralCode	29.83	18.93	56.31

Currently trained on 100K / 2.05M (∼1/20) of data

Status: Experimental

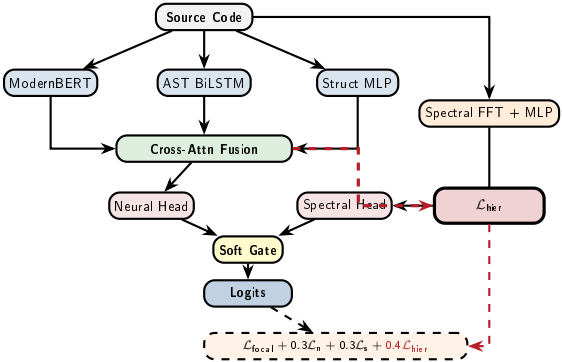
- T1: Competitive with ModernBERT (29.83 vs 30.61)
- T3: Competitive (56.31 vs 61.65)
- T2: Below — need family-aware loss

→ **Using only 1/20 of training data!**

AICD T1 OOD Collapse

Val 99.5% → Test 29.8% **across ALL methods**. 4-split progressive OOD (unseen langs + domains) is fundamentally hard.

HierTreeCode — SpectralCode + Hierarchical Affinity Loss



Hierarchical Affinity Loss

Family Tree (adapts per benchmark):

Qwen	$\rightarrow$ {Qwen1.5, Nxcode}
Meta	$\rightarrow$ {Llama3.1, CodeLlama}
Google	$\rightarrow$ {Gemma, Gemini}

Batch-hard triplet:

$$[\, d_{\textbf{pos}}^{\max} - d_{\textbf{neg}}^{\min} + \alpha \,]_+$$

d: cosine distance on ℓ_2-norm $h_{\textbf{neural}}$
$\alpha = 0.3$ margin

Effect: Forces same-family closer, different-family further apart.

Difference from SpectralCode

SpectralCode = this diagram **without** the red $\mathcal{L}_{\textbf{hier}}$ branch. Everything else is identical.

HierTreeCode: Preliminary Results + Expected Impact

CoDET-M4 Author F1 (✓ Done)

	Author F1	Δ vs UniX
UniXcoder	66.33	—
SpectralCode	69.82	+3.49
HierTreeCode	**70.55**	**+4.22**

Per-generator F1 (HierTreeCode):

Human	GPT	Llama	CodeLlama
0.87	0.92	0.67	0.74

Qwen1.5	Nxcode	Macro
0.44 (+.03)	**0.59**	**70.55**

Why HierTreeCode Will Help Further

- **DROID T3/T4**: Family hierarchy (human → refined → machine → adversarial)
- **AICD T2** (12-class): Groups Google (Gemma+Gemini), Chinese AI (DeepSeek+Qwen)
- **AICD T3**: Adversarial ≈ machine origin
- Family tree adapts per benchmark automatically

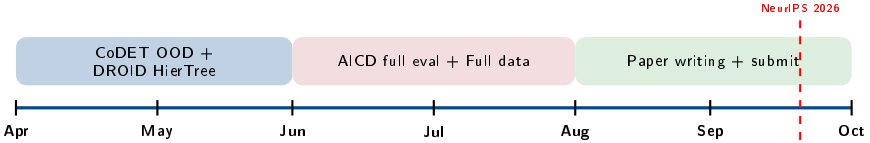
Hierarchical Affinity Loss

$$\mathcal{L}_{\text{hier}} = \frac{1}{|\mathcal{A}|} \sum_i [d_{i,p^*} - d_{i,n^*} + \alpha]_+$$

p^*: farthest same-family (hard positive)
n^*: closest diff-family (hard negative)
Cosine distance on ℓ_2-normalized h_{neural}

Full Evaluation Matrix

	CoDET-M4 ACL'25	DROID EMNLP'25	AICD EACL'26
Binary	✓ 99.06%	✓ 88.77%^{w} T3	🕓 29.83% T1
Attribution	✓ 70.55% (6-cls)	—	🕓 T2 (12-cls)
Refined	—	✓ 88.77%^{w} T3	🕓 T3
Adversarial	—	✓ 88.02%^{w} T4	🕓 T3
OOD Lang	🕓 LOO	—	🕓 3→9 lang



Contributions & Summary

Technical Contributions

1. **SpectralCode**: Multi-stream (token + AST + structural + spectral FFT) with cross-attention + gating — *SOTA on CoDET-M4, beats DroidDetect-Base on DROID*
2. **HierTreeCode**: First use of LLM genealogy as structured prior — encodes parent-child as embedding distance constraints via batch-hard triplet mining
3. **Unified 3-benchmark evaluation**: First work covering CoDET-M4 + DROID + AICD — binary + attribution + OOD + adversarial in one method
4. **Scaling insight**: Competitive results using only 1/10–1/20 of training data — full data expected to close remaining gaps

Results So Far

	CoDET-M4	DROID	AICD
Binary	99.06% (+0.41 vs UniX)	88.77%^{w} (>DroidDet-B)	⏲ ongoing
Author	70.55% (+4.22 vs UniX)	—	⏲ T2
Data used	100K/500K	100K/1.06M	100K/2.05M

CoDET-M4: 10 Methods Explored — What Worked and What Didn't

Full Leaderboard (Author 6-class F1)

#	Method	Author F1	Key Idea
1	**HierTreeCode**	**70.55**	Family tree loss
2	RAGDetect	70.46	Retrieval-augmented
3	BiScopeCode	70.20	MLM memorization
4	TTACode	70.20	Test-time adapt
5	MultiGranCode	70.19	CharCNN fusion
6	GroupDRO	70.17	Worst-group opt
7	SelfDistillCode	70.14	EMA distillation
8	ProtoCon	70.13	Prototype + unif
9	SpectralCode	69.82	FFT spectral
10	GraphStyleCode	69.71	GAT on AST
—	EAGLECode	62.89	DANN (failed)

DANN Catastrophically Fails

Gradient reversal forces generator-invariant features — this *directly opposes* attribution.
Qwen1.5 F1 drops to **0.198** (random).
Lesson: Domain adversarial $\neq$ attribution.

Key Insights

- All methods cluster in 69.7–70.6% author F1
- Qwen1.5/Nxcode confusion is THE bottleneck
- GitHub source ↓21% vs CodeForces
- GAT on AST provides no benefit
- Spectral FFT is surprisingly effective

AICD + DROID: 13 Deep Methods — Cross-Benchmark Insights

Leaderboard (5 completed methods)

#	Method	AICD	DROID	Avg
1	SpectralCode	.350	.847	**.549**
2	TokenStat	.330	**.852**	.539
3	MoE-Code	.324	.849	.534
4	SlotCode	.329	.839	.533
5	DomainMix	.286	.716	.458

Methods That Failed

CausAST: Ortho → 0 (worst) | **OSCP**: Whitening dominates | **AST-IRM**: Penalty explodes to 5000

Cross-Benchmark Insights

1. **AICD T1 OOD unsolved**: Val 99.5% → Test 30% for ALL methods
2. **Spectral features transfer best**: Wins overall despite simplicity
3. **Token stats are powerful**: Entropy, burstiness, Yule-K beat complex architectures on DROID
4. **Mixup helps OOD**: DomainMix achieves best AICD T1 via embedding mixup
5. **Only 1/10–1/20 data used**: Full data → higher scores expected

Thank You

Questions & Discussion

CoDET-M4 (ACL Findings'25) · DroidCollection (EMNLP'25) · AICDBench (EACL'26)

Code and experiments available upon request